Huawei MG-ACC-46 Access Node QoS / Congestion Event (Microbursts, interface tail drops, QoS queue starvation) Troubleshooting and Resolution SOP

Node: MG-ACC-46 | **Vendor:** Huawei | **Severity:** Minor | **Alarm Code:** ALARM_7490

Description

The failure domain of QoS / Congestion Event (Microbursts, interface tail drops, QoS queue starvation) on the Huawei MG-ACC-46 Access node is primarily attributed to the inherent complexities of Quality of Service (QoS) policies and their interaction with the underlying packet forwarding mechanisms. In the context of this specific node and vendor equipment, the root cause of this issue can be linked to the misconfiguration or inadequate tuning of QoS policies, leading to the formation of microbursts and subsequent interface tail drops. Furthermore, the possibility of QoS queue starvation cannot be ruled out, where high-priority traffic is excessively consuming available bandwidth, thereby causing the queues of lower-priority traffic to become depleted. This, in turn, can trigger the interface to drop packets, exacerbating the congestion event. The Huawei MG-ACC-46 Access node's reliance on the Huawei proprietary QoS architecture, which integrates various components such as the Enhanced QoS (EQoS) mechanism and the Traffic Policing (TP) function, further complicates the troubleshooting process.

Pre-requisite Checks

- Ensure that the remote access to the MG Road Core site is established and maintained.

- Verify the status of redundant links and confirm that they are operational and not contributing to the congestion event.

- Check the power supply and environmental conditions at the site to rule out any external factors that may be influencing the node's performance.

- Confirm that the necessary personnel and equipment are available to perform the troubleshooting and resolution procedures.

Potential Root Causes

Cause	Probability
Misconfigured or inadequately tuned QoS policies	High
QoS queue starvation due to excessive high-priority traffic	Medium
Interface hardware or firmware issues	Low
Misconfigured or malfunctioning Enhanced QoS (EQoS) mechanism	Medium
Malfunctioning Traffic Policing (TP) function	Low

Diagnostic Commands & Expected Outputs

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Command: display qos policy, Expected Output: Display QoS policy configuration and statistics., Description: This command will provide an overview of the QoS policy configuration, including the policing rates, shaping rates, and queue configurations.

Command: display interface, Expected Output: Display interface configuration and statistics., Description: This command will provide an overview of the interface configuration, including the interface type, IP address, and statistics such as RX and TX packets and bytes.

Command: display qos queue statistics, Expected Output: Display QoS queue statistics., Description: This command will provide detailed statistics about the QoS queues, including the number of packets and bytes in each queue, as well as the queue drop rates.

Command: display eqos statistics, Expected Output: Display EQoS statistics., Description: This command will provide detailed statistics about the EQoS mechanism, including the number of packets processed and dropped, as well as the EQoS queue drop rates.

Command: display tp statistics, Expected Output: Display TP statistics., Description: This command will provide detailed statistics about the TP function, including the number of packets processed and dropped, as well as the TP queue drop rates.
```

Resolution Steps

Step 1: Step: Verify and adjust QoS policy configuration, Command: undo qos policy, Description: This command will remove the existing QoS policy configuration. After removing the policy, proceed to reconfigure the QoS policy with the correct policing rates, shaping rates, and queue configurations.

Step 2: Step: Adjust queue configurations, Command: queue-configuration interface, Description: This command will allow you to adjust the queue configurations, including the queue sizes, priority levels, and queue drop rates.

Step 3: Step: Verify and adjust EQoS configuration, Command: undo eqos, Description: This command will remove the existing EQoS configuration. After removing the EQoS, proceed to reconfigure the EQoS with the correct settings.

Step 4: Step: Verify and adjust TP configuration, Command: undo tp, Description: This command will remove the existing TP configuration. After removing the TP, proceed to reconfigure the TP with the correct settings.

Post-Check Verification

```
Command: display qos policy, Expected Output: Display QoS policy configuration and statistics., Description: This command will verify that the QoS policy configuration is correct and that the queues are properly configured.

Command: display interface, Expected Output: Display interface configuration and statistics., Description: This command will verify that the interface configuration is correct and that the interface is operational.

Command: display qos queue statistics, Expected Output: Display QoS queue statistics., Description: This command will verify that the QoS queues are properly configured and that the queue drop rates are acceptable.

Command: display eqos statistics, Expected Output: Display EQoS statistics., Description: This command will verify that the EQoS mechanism is operating correctly and that the EQoS queue drop rates
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are acceptable.

Command: display tp statistics, Expected Output: Display TP
statistics., Description: This command will verify that the TP
function is operating correctly and that the TP queue drop rates are
acceptable.
```

Roll-back Procedures

- Step: Restore original QoS policy configuration, Command: qos policy, Description: This command will restore the original QoS policy configuration.

- Step: Restore original EQoS configuration, Command: eqos, Description: This command will restore the original EQoS configuration.

- Step: Restore original TP configuration, Command: tp, Description: This command will restore the original TP configuration.